

Jose E. Rodríguez-Ríos

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joserico00.github.io

EDUCATION

Ph.D., Software Engineering , University of Puerto Rico - Mayagüez	Aug 2026 - Present
M.S., Computer Science , University of Puerto Rico - Río Piedras GPA: 4.0/4.0	Aug 2022 - May 2025
B.S., Computer Science , University of Puerto Rico - Río Piedras Minor in Cybersecurity	Aug 2015 - Dec 2020

TECHNICAL SKILLS

Programming languages: Python, C, C++, Assembly

AI and HPC: PyTorch, Hugging Face, TensorFlow, transformers, SLURM, GPU computing, Jupyter

Cybersecurity: Nmap, OpenVAS, Ghidra, Wireshark, Scapy, Metasploit, Hashcat, Burp Suite, Autopsy

Systems and DevOps: Linux, Docker, GitLab CI/CD, Git, Elasticsearch, Flask, Streamlit, SQLite, MySQL, OpenVPN

EXPERIENCE

Research Intern, Argonne National Laboratory, Lemont, IL Jun 2026 - Aug 2026

- Built DNA classifiers with **PyTorch**, **Hugging Face**, and **DNABERT** on **Perlmutter**, **Polaris**, and **Aurora**. Compared **6-mer/BPE** tokenization: **81%/80%** accuracy and **195/34 tokens** per sequence.
- Led a **10-student** team as project lead: scoped a **10-12-hour** project, wrote **Jupyter/SLURM** lessons and **CPU-versus-GPU** experiments, and coached final presentations; its fine-tuned **DNABERT** reached **91.7%** evaluation accuracy on **CTCF** data.

Volunteer Software Developer, Abeyno Coop solar cooperative, Salinas, PR 2024 - Present

- Built Solar Fleet Reporter, a **1,700-line Streamlit** app with **88 automated checks** that pulls energy data from **Sol-Ark**, **FranklinWH**, and **sonnen** portals, applies net-metering billing rules, and exports **CSV/Excel** reports and printable homeowner statements.

DevOps Intern, Lawrence Livermore National Laboratory, Livermore, CA Jun 2024 - Aug 2024

- Built a containerized **Elasticsearch** store for **Ghidra BSIm** binary features so researchers can query malware-like samples at scale, with **GitLab CI/CD** for deployment.
- Deployed a **Docker**-hosted **Ollama (Llama 3)** service for the **PaperQA** PDF-reading application.

Graduate Research and Teaching Assistant, UPR Río Piedras, San Juan, PR Aug 2022 - May 2025

- Developed **NVSRCO**, an open-source vulnerability scanner using **Python**, **Nmap**, and **OpenVAS**, with **CSV** reporting, a **Dash** dashboard, and **WireGuard** links to remote networks.
- Led Operating Systems, Computer Architecture, and Cybersecurity labs (**Linux**, **Ghidra**, **Wireshark**, **Scapy**); authored **Capture the Flag (CTF)** challenges in reverse engineering, cryptography, web, and forensics for about **60 Puerto Rico high-school students**.

Cybersecurity Intern, GM Sectech, San Juan, PR Jun 2022 - Aug 2022

- Built a **Python/Twilio** smishing (SMS phishing) simulator for awareness training; collaborated with penetration testers on credential-theft and domain-compromise scenarios; trained on **Metasploit**, **Hashcat**, and **Burp Suite**.

SELECTED PROJECTS (full list at joserico00.github.io)

Machine Learning for Security

- Trained a **TensorFlow CNN** on **Maling (9,401 images, 25 malware families)**, reaching **97.7%** validation accuracy.
- Built an **IoT-23** traffic pipeline with capture-aware splits to prevent data leakage.

yt2ogg

- Built a desktop/CLI audio tool with a pure **NumPy** beat tracker to find a track's beat grid and render gapless loops.
- Packaged standalone **Windows** executables with self-tests.

CYBERSECURITY AWARDS

BSides Puerto Rico CTF (2024): 1st place of 30 teams; reverse engineering, cryptography, exploitation.

Tracer Fire Blue Team, ACSAC (2019): 2nd place team of 40 participants; SIEM/IDS, Autopsy, Ghidra.